הייטקורס
לחשוב בגדול

בשיתוף
JOHN BRYCE
#1 בלימודי הייטק
a matrix company

מהבסיס - כולל תרגול

Advanced GenAI & Agentic AI for Developers

מפיתוח יישומי GenAI לבניית מערכות Agentic AI מתקדמות

עולם הבינה המלאכותית מתקדם במהירות ממערכות שמייצרות תוכן למערכות חכמות שמסוגלות לתכנן, לקבל החלטות, להפעיל כלים ולבצע משימות מורכבות. המעבר מ Generative AI לAgentic AI משנה את האופן שבו מפתחים מתכננים מערכות תוכנה, מנהלים תהליכים ומשלבים יכולות AI במוצרים ובסביבות ארגוניות.

הקורס מיועד למפתחים המעוניינים להתקדם• לחזית הפיתוח של מערכות AI ולהכיר את הכלים, התשתיות ודפוסי העבודה המשמשים לבניית סוכני AI מתקדמים. במהלך הקורס נלמד כיצד לפתח מערכות סוכנות מבוססות מודלי שפה, לחבר אותן למידע ולכלים חיצוניים, לנהל תהליכים מרובי שלבים ולבנות פתרונות מאובטחים, יעילים וניתנים לניטור.

הקורס משלב היכרות מעמיקה עם Frameworks מובילים, ארכיטקטורות מתקדמות וכלים מעשיים לפיתוח מערכות Agentic AI המותאמות לעבודה בסביבות אמת.

סילבוס זה הוא בסיסי בהתאם לידוע לנו בתחום בשלב זה.

תכני הקורס יתעדכנו סמוך לקורס בהתאם לחידושים בתעשייה

Prerequisities:

Development Environment:

1. Node.js
2. npm
3. TypeScript
4. Docker / Docker Desktop

LLM and Agent Foundations

Topics

- What an LLM is and how it works at the application level
- The difference between Traditional ML and Generative AI
- Capabilities and limitations of language models - compare latest used models (astra, fable, gemini)
- What an Agent is and how it differs from a regular Chat or LLM Application
- Introduction to Tool Use and Function Calling

Prompt Engineering for Developers

Topics

- Writing clear instructions for the model, Role Assignment & Basic Guardrails
- Explore prompts providers - fetching well designed prompts.
- Output Structuring and producing valid JSON from a model response

Lab: Building a component that takes free text from the user and converts it into a structured command:

- Defining the output format
- Building the prompt
- Testing different input scenarios
- Refining the prompt based on failures
- Returning a structured result in code

MCP – Connecting Models to External Tools

Topics

- What the Model Context Protocol is and why it was created
- The difference between a regular Tool and an MCP Server
- Local MCP vs. Remote MCP
- Exposing external actions to the model
- Basic security considerations
- Managing context when there are many tools

Core Exercise

Building and connecting a basic MCP Server:

- Defining an external action as a Tool
- Exposing the action via MCP
- Connecting an Agent to MCP
- Running an action through the model and testing basic scenarios and errors

Skills & Rules – Building Reusable Capabilities and instructions

Topics

- What Skills and Rules are, and how they differ from Tools and Agents
- When to use a Skill vs a Rule
- Basic structure and principles for writing clear, reusable Skills and Rules
- Progressive Disclosure and using scripts/resources inside Skills
- Combining Skills and Rules for consistent project behavior and reusable workflows

Subagents - Delegating Work to Specialized Agents

Topics

- What subagents are and when to use them for specialization, context isolation, and parallel work
- How a parent agent delegates tasks and receives results back from subagents
- Defining custom subagents and their instructions, tools, models, and responsibilities
- Platform conventions for declaring agents, such as AGENTS.md-style project instructions

Agent Instructions and Evaluations

Topics

- Using AGENTS.md for project-level agent instructions and conventions
- Tracing agent execution, tool calls, and workflow behavior
- Evaluating agent outputs and defining success criteria
- Cost, token usage, and latency analysis
- Failure analysis, debugging, and regression testing

RAG, Vector Search, and GraphRAG

Topics

- What RAG is and why a model needs external knowledge
- Embeddings and Vector Search
- Chunking and the basic structure of a RAG pipeline
- Injecting context into a model's response and common RAG limitations
- GraphRAG: when vector search isn't enough - representing knowledge as an entity-relationship graph

- The difference between retrieval by semantic similarity (Vector) and retrieval by structure and relationships (Graph)
- Hybrid combination of Vector RAG and GraphRAG in the same pipeline

Core Exercise

Building a small knowledge base and using it for grounded answers:

- Loading information and splitting it into chunks
- Creating embeddings and performing vector retrieval
- Building a basic entity graph from the same information and retrieving through it
- Combining both retrieval sources in the prompt and returning an answer with a source, or a message that there isn't enough information

LLM Wiki – Andrej Karpathy's Insights

Topics

- How to create knowledge base to your development agent.
- Maintain documentation.
- What an LLM Wiki is and how it differs from traditional RAG.
- Using the LLM to transform raw information into a structured, persistent knowledge base.
- Organizing knowledge into interconnected pages, topics, and references instead of relying only on document chunks
- Using the Wiki as a reusable knowledge layer for Agents, Skills, and other AI workflows
- The relationship between LLM Wiki, RAG, and GraphRAG - when to retrieve raw information and when to use synthesized knowledge

Wrap it up:

Assembling an end-to-end agentic process that combines a Skill, an MCP connection, and knowledge retrieval (Vector + Graph) into one complete scenario